

Trainings ▾

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery last.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries and air starter to prevent accidental starting.
- Drain the lubricating oil system. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>).
- Remove the oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).
- Remove the fan and fan spacer(s). Refer to Procedure 008-040 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-040.html>).
- Remove the fan belt. Refer to Procedure 008-002 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>).
- Remove the alternator belt and alternator. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>).

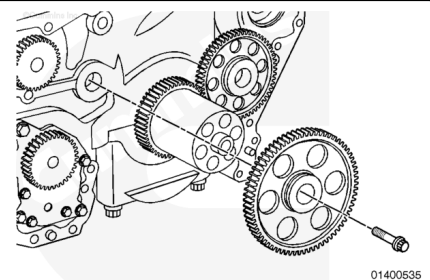
- Remove the fan belt tensioner assembly. Refer to Procedure 008-087 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-087-tr.html>).
- Remove the vibration damper. Refer to Procedure 001-052 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-052.html>).
- Remove the front crankshaft seal. Refer to Procedure 001-023 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-023.html>).
- Remove the front gear cover. Refer to Procedure 001-031 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-031.html>).

Remove

The bolt-in idler shafts have a flange that require the shaft, the idler gear, and the thrust washer to be removed as an assembly.

Remove the mounting capscrew from the idler gear.

Remove the gear assembly using two pry bars to gently pry the gear and the shaft from the block.



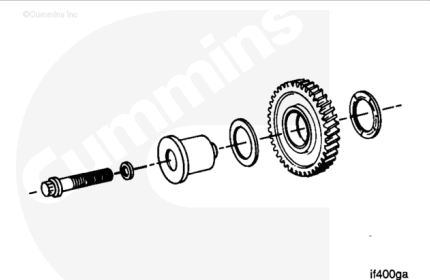
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



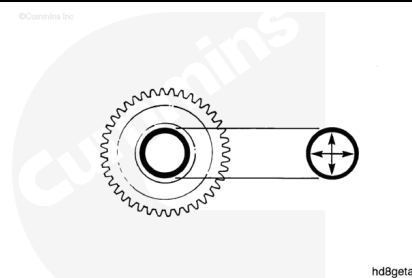
Use solvent. Clean the idler gear.

Dry with compressed air.

Check all parts for damage.

Measure the inside diameter of the idler gear bushing.

Idler Gear Bushing Inside Diameter		
mm		in
47.639	MIN	1.8756
47.663	MAX	1.8765



Measure the outside diameter of the idler gear bushing.

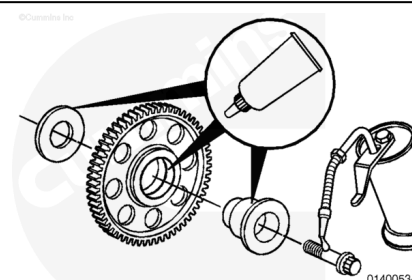
Idler Gear Bushing Outside Diameter		
mm		in
50.900	MIN	2.004
50.927	MAX	2.005

Note : The bushing in the gear is precision bored after installation. If machining capability is **not** available, replace the bushing and the gear as an assembly.

Install

⚠ CAUTION ⚠

The chamfer on the thrust washer must be turned toward the gear.

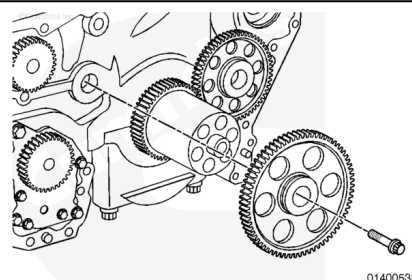


Use Lubriplate™ 105, or equivalent. Lubricate the gear bushing, shaft, and the thrust washer.

Use clean 15W-40 engine oil to lubricate the capscrew.
Assemble the parts as shown.

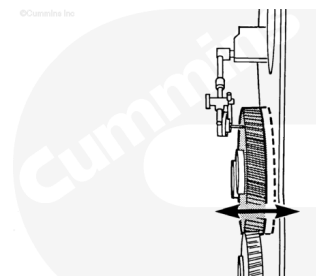
Use the capscrew to pull the shaft into the bore. Tighten the capscrew.

Torque Value: 277 n·m [204 ft-lb]



Use a dial indicator to measure the idler gear end clearance.

Water Pump - Idler Gear End Clearance		
mm		in
0.05	MIN	0.002
0.17	MAX	0.007



If the clearance is **not** within specifications, check for foreign material between the parts or check for proper location of the thrust washers. Oversize washers are available.

Check the idler gear backlash. Refer to Procedure 001-055 (</qs3/pubsys2/xml/en/procedures/89/89-001-055.html>).

Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the front gear cover. Refer to Procedure 001-031 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-031.html>).
- Install the front crankshaft seal. Refer to Procedure 001-023 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-023.html>).
- Install the vibration damper. Refer to Procedure 001-052 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-052.html>).
- Install the fan belt tensioner assembly. Refer to Procedure 008-087 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-087-tr.html>).
- Install the alternator. Refer to Procedure 013-001 in Section 13 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>).
- Install the alternator belt. Refer to Procedure 013-005 in Section 13 (</qs3/pubsys2/xml/en/procedures/89/89-013-005.html>).
- Install the fan belt. Refer to Procedure 008-002 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>).
- Install the fan spacer(s) and fan. Refer to Procedure 008-040 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-040.html>).
- Install new oil filters. Refer to Procedure 007-013 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-013.html>).
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).
- Install the oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).

- Fill the lubricating oil system. Refer to Procedure 007-037 in Section 7 (/qs3/pubsys2/xml/en/procedures/89/89-007-037.html).
- Connect the batteries.
- Operate the engine to normal operating temperature. Check for leaks.

Last Modified: 12-Apr-2022
